CIE INSIDER | REVISION BOOKLET

1.2 Motion

Cambridge IGCSE Physics (0625) | Syllabus 2026–2028

Tier: Core and Extended

Name: _____ Class: _____ Date: _____

How to use this booklet

1. Rate every syllabus statement Red, Amber or Green now, then again after you have worked through the booklet.
2. Draw the diagrams in Section 4 yourself. Drawing and labelling them is a skill the exam tests, so a blank box here is deliberate.
3. Answer the practice questions before you look at the mark scheme. Your teacher has the mark scheme and model diagrams.

Section 1: Syllabus checklist

Blue = Core | Amber = Extended (Supplement)

#	Syllabus statement (verbatim)	Tier	R	A	G
1	Define speed as distance travelled per unit time; recall and use the equation $v = s/t$	Core			
2	Define velocity as speed in a given direction	Core			
3	Recall and use the equation: average speed = total distance travelled / total time taken	Core			
4	Sketch, plot and interpret distance–time and speed–time graphs	Core			
5	Determine, qualitatively, from given data or the shape of a distance–time graph or speed–time graph when an object is: (a) at rest (b) moving with constant speed (c) accelerating (d) decelerating	Core			
6	Calculate speed from the gradient of a straight-line section of a distance–time graph	Core			
7	Calculate the area under a speed–time graph to determine the distance travelled for motion with constant speed or constant acceleration	Core			
8	State that the acceleration of free fall g for an object near to the surface of the Earth is approximately constant and is approximately 9.8 m/s^2	Core			
9	Define acceleration as change in velocity per unit time; recall and use the equation $a = \Delta v/\Delta t$	Ext			
10	Determine from given data or the shape of a speed–time graph when an object is moving with: (a) constant acceleration (b) changing acceleration	Ext			
11	Calculate acceleration from the gradient of a speed–time graph	Ext			
12	Know that a deceleration is a negative acceleration and use this in calculations	Ext			
13	Describe the motion of objects falling in a uniform gravitational field with and without air/liquid resistance, including reference to terminal velocity	Ext			

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Section 2: Key definitions

Term	Definition
Speed	Distance travelled per unit time (scalar)
Velocity	Speed in a given direction (vector)
Average speed	Total distance travelled divided by total time taken
Acceleration of free fall (g)	The acceleration of an object falling freely near the Earth's surface; approximately constant, $g \approx 9.8\ m/s^2$
Acceleration	Change in velocity per unit time; $a = \Delta v / \Delta t$ (vector)
Deceleration	A negative acceleration: the object is slowing down
Terminal velocity	The constant velocity a falling object reaches when the air (or liquid) resistance on it equals its weight, so the resultant force is zero

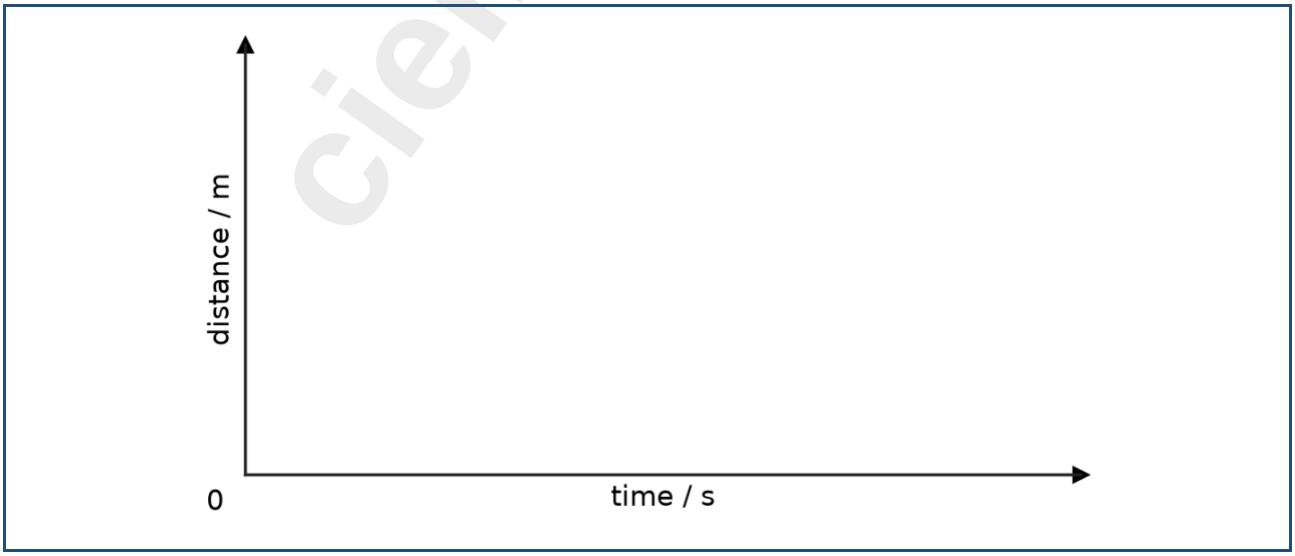
Section 3: Formulae and equations

Quantity	Equation	Symbols and units
Speed	**$v = s / t$**	v = speed (m/s), s = distance (m), t = time (s)
Average speed	**average speed = total distance / total time**	m/s = m / s
Acceleration	**$a = \Delta v / \Delta t$**	a = acceleration (m/s^2), Δv = change in velocity (m/s), Δt = time taken (s)

Section 4: Diagrams to draw

4.1 Distance–time graph Core

Task: Sketch a distance–time graph for a journey in four stages, in this order: A at rest, B moving at constant speed, C accelerating, D decelerating. Label each stage with its letter.



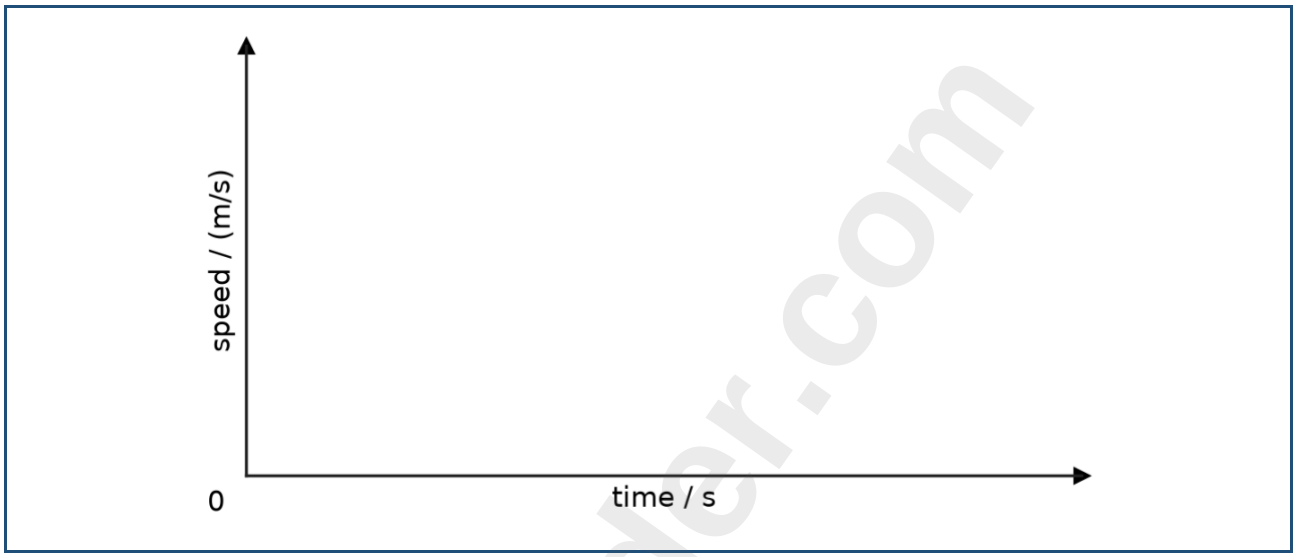
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Check your diagram has:

- ☐ A is a horizontal line (gradient zero)
- ☐ B is a straight line sloping upwards
- ☐ C curves upwards (gradient increasing)
- ☐ D curves over (gradient decreasing)
- ☐ Stages labelled A, B, C, D

4.2 Speed–time graph Core

Task: Sketch a speed–time graph in three stages: A constant speed, B constant acceleration, C (Extended) changing acceleration. Shade the area that gives the distance travelled during stage B.



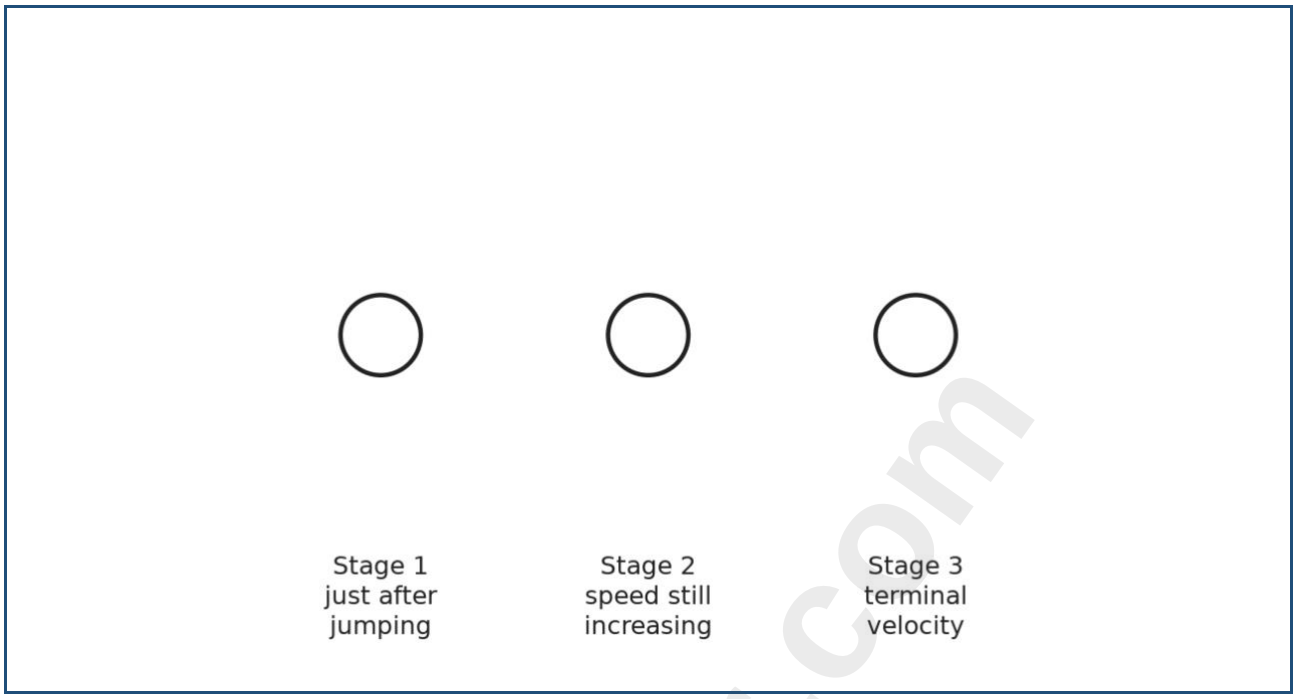
Check your diagram has:

- ☐ A is a horizontal line above the time axis
- ☐ B is a straight line sloping upwards
- ☐ Extended: C is a curve (gradient changing)
- ☐ Area under B shaded and labelled 'distance'

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4.3 Forces on a falling object Extended

Task: The circles show a skydiver at three stages of a fall before the parachute opens. Draw and label the weight and the air resistance on each. Make the arrow lengths show how the forces compare.



Check your diagram has:

- ☐ Weight arrow pointing down, the same length at every stage
- ☐ Air resistance arrow pointing up, getting longer from stage 1 to stage 3
- ☐ Stage 3: air resistance arrow equal in length to weight
- ☐ Both forces labelled

Section 5: Worked examples

Worked Example 1 Core

A distance–time graph is a straight line from (0 s, 0 m) to (5.0 s, 35 m). Calculate the speed of the object.

speed = gradient of the distance–time graph

speed = change in distance / change in time = (35 − 0) m / (5.0 − 0) s

speed = 35 / 5.0

speed = 7.0 m/s

Worked Example 2 Core

A speed–time graph is a straight line from (0 s, 0 m/s) to (8.0 s, 20 m/s). Calculate the distance travelled.

distance = area under the speed–time graph

The shape is a triangle: area = ½ × base × height

distance = ½ × 8.0 s × 20 m/s

distance = 80 m

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Worked Example 3 Extended

A speed–time graph is a straight line from (2.0 s, 6.0 m/s) to (7.0 s, 21.0 m/s). Calculate the acceleration.

acceleration = gradient of the speed–time graph

$a = \Delta v / \Delta t = (21.0 - 6.0)\ m/s\ /\ (7.0 - 2.0)\ s$

$a = 15.0 / 5.0$

$a = 3.0\ m/s^2$

Section 6: Multiple choice

Paper 1 and Paper 2 style. Choose one answer, A, B, C or D. 8 marks, one for each question.

1. Which statement defines velocity?

A distance travelled per unit time
B speed in a given direction
C change in velocity per unit time
D change in speed

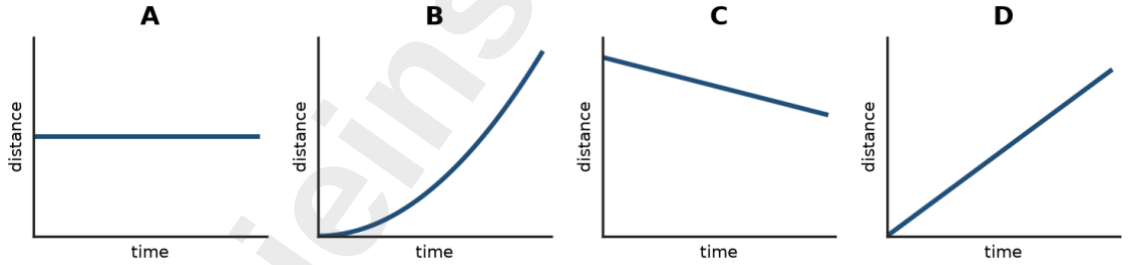
Answer: _____

2. A cyclist rides 3.0 km uphill in 10 minutes, then 3.0 km downhill in 5.0 minutes. What is the average speed for the whole journey?

A 5.0 m/s
B 6.7 m/s
C 7.5 m/s
D 10 m/s

Answer: _____

3. The speed–time graph of a car is a horizontal line above the time axis. Which distance–time graph shows the same motion?



Answer: _____

4. A car starts from rest and accelerates uniformly to 20 m/s in 8.0 s. What distance does it travel in this time?

A 2.5 m
B 20 m
C 80 m
D 160 m

Answer: _____

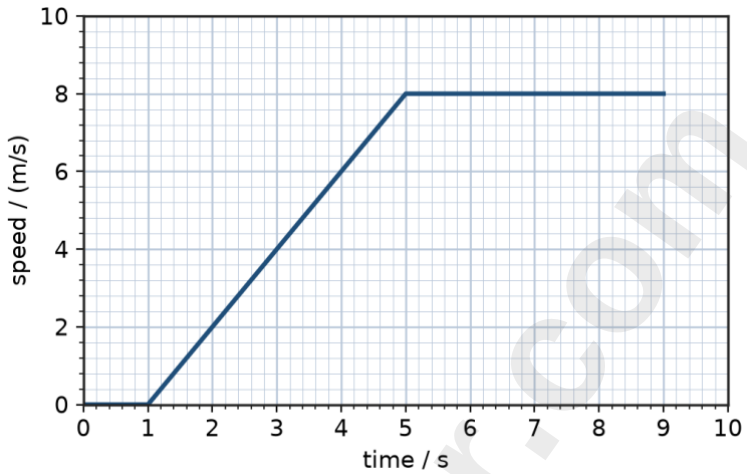
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5. A ball is dropped near the Earth's surface. Air resistance is negligible. What happens to the acceleration of the ball as it falls?

A It stays constant at about 9.8 m/s².
B It increases as the ball speeds up.
C It decreases as the ball falls.
D It decreases to zero when the ball reaches terminal velocity.

Answer: _____

6. Extended The graph shows the speed of an athlete at the start of a race. What is the acceleration of the athlete while the speed is increasing?



A 0.89 m/s²
B 1.6 m/s²
C 2.0 m/s²
D 16 m/s²

Answer: _____

7. Extended A ball hits a wall at 6.0 m/s and bounces straight back at 4.0 m/s. It is in contact with the wall for 0.20 s. What is the size of its average acceleration while in contact with the wall?

A 2.0 m/s²
B 10 m/s²
C 25 m/s²
D 50 m/s²

Answer: _____

8. Extended A skydiver is falling at terminal velocity. Which statement is correct?

A The resultant force is zero and the speed is constant.
B The speed is still increasing, but slowly.
C The air resistance is greater than the weight.
D No forces act on the skydiver.

Answer: _____

Section 7: Structured questions

Total: 41 marks. Show your working in every calculation.

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1. State the value of the acceleration of free fall g near the Earth's surface. Include its unit. [1]

2. Explain the difference between speed and velocity. [2]

3. A car travels a total distance of 150 km in 2.5 hours. Calculate the average speed of the car in m/s. [3]

4. A speed–time graph for an object is a horizontal line at 12 m/s from t = 0 to t = 6.0 s. It is then a straight line rising from 12 m/s to 30 m/s between t = 6.0 s and t = 12 s.

(a) Calculate the total distance travelled in the 12 s. [3]

(b) Extended Calculate the acceleration between t = 6.0 s and t = 12 s. [2]

5. Describe the shape of the line on a distance–time graph for an object that is decelerating. Explain what the shape tells you about the object's speed. [2]

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6. Extended Define acceleration. An object's velocity changes from 4.0 m/s to 22.0 m/s in 3.0 s. Calculate its acceleration. [3]

7. Extended A skydiver jumps from a plane.

(a) Describe how the skydiver's speed would change during the fall if there were no air resistance. [1]

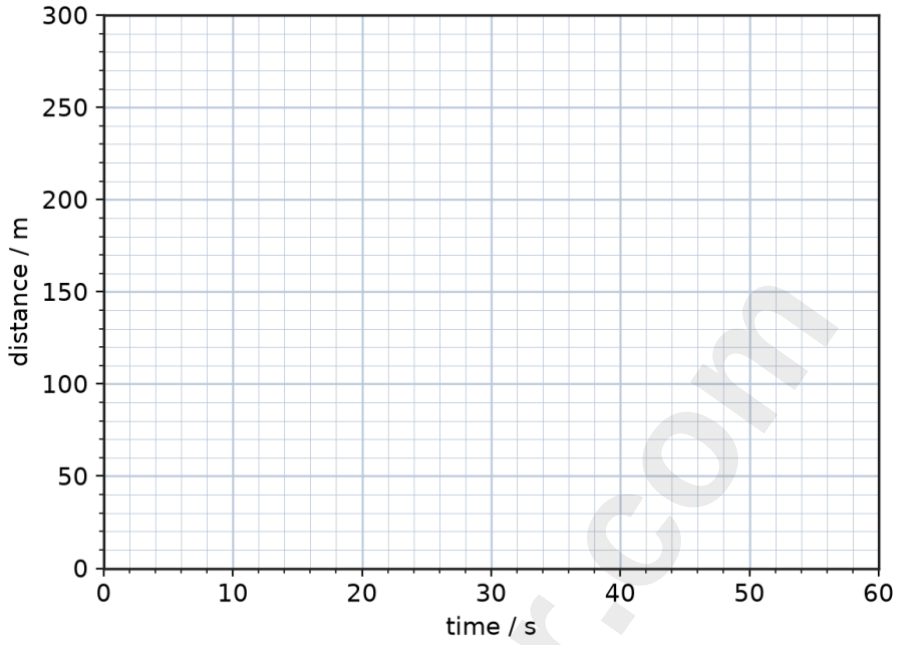
(b) With air resistance, the skydiver reaches terminal velocity before opening the parachute. Explain, in terms of the forces acting, why this happens. [4]

(c) The skydiver then opens the parachute. Describe and explain the motion of the skydiver from the moment the parachute opens until the skydiver is falling at a steady speed again. [3]

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8. A cyclist's distance from home is measured every 10 s. The table shows the results.

time / s	0	10	20	30	40	50	60
distance / m	0	60	120	180	180	220	260



(a) On the grid, plot the points and join them with straight lines. [3]

(b) State what the cyclist is doing between t = 30 s and t = 40 s. [1]

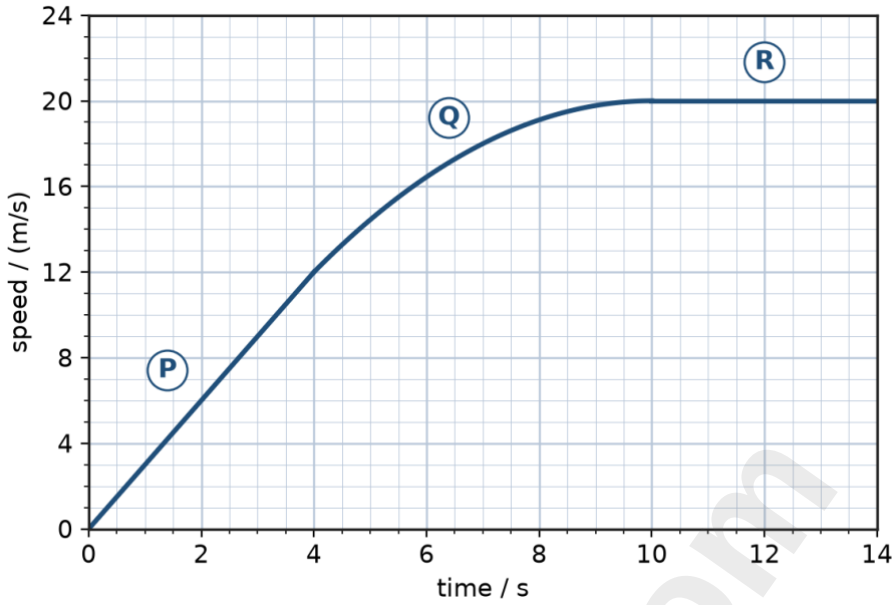
(c) Calculate the cyclist's speed during the first 30 s. [2]

(d) Calculate the average speed for the whole 60 s. [2]

9. **Extended** A cyclist moving at 8.0 m/s brakes and stops in 2.5 s. Calculate the acceleration of the cyclist and state what the sign of your answer tells you. [3]

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10. Extended The speed–time graph shows the motion of a car in three sections, P, Q and R.



(a) State which section shows constant acceleration. [1]

(b) Describe the motion of the car in section Q. [2]

(c) Calculate the acceleration of the car in section P. [2]

(d) State the acceleration of the car in section R. [1]

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Section 8: Examiner tips

Where a tip comes from a Cambridge examiner report for 0625, the exam series and paper are given in brackets.

- Use g = 9.8 m/s², not 10. The syllabus has required 9.8 since 2023, and the June 2023 Paper 4 reports repeat this for all three variants. (June 2023, Paper 4)
- Speed versus velocity: a definition of velocity must include the word direction.
- Gradient or area? The gradient of a distance–time graph is speed. The gradient of a speed–time graph is acceleration. The area under a speed–time graph is distance. Reports keep finding these swapped, for example working out an acceleration or deceleration from the area under the graph. (June 2024 and Nov 2025, Paper 22)
- Split the area under a speed–time graph into rectangles and triangles, and work each one out separately. Treating a triangle as a rectangle (forgetting the ½) was the most common error on this question in March 2021, and trapezium-shaped areas gained full credit only from stronger candidates in Nov 2022. (March 2021 and Nov 2022, Paper 3)
- When you describe a section of a graph, name the motion in full: 'constant speed', not just 'constant'. 'Constant' on its own gained no credit. (Nov 2025, Paper 31)
- Define acceleration as 'change in velocity per unit time' or 'rate of change of velocity'. Do not combine the two. (June 2024, Paper 41)
- With no air resistance, a falling object's acceleration stays constant at g. Candidates have lost marks believing it increases, or decreases, as the object falls. (Nov 2024, Papers 11 and 21)
- Terminal velocity needs the whole chain: speed increases, so air resistance increases, so the resultant force decreases, so the acceleration decreases, until air resistance equals weight. Few candidates described the forces before terminal velocity is reached. (June 2021, Paper 4)
- Deceleration is a negative acceleration. In a calculation, keep the minus sign: a = −3.2 m/s² means slowing down.

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